

Procedure	Function
May or may not return a value	Must return a value.
①. Not return a value	
②. Uses out parameters to return values.	Uses Return Statement.
③. Used to perform actions.	Used to Calculate and return a value.
④. Return type not mandatory.	Return type mandatory.
⑤. Called using Exec or from PL/SQL block.	Can be called in SQL statements.

★ Trigger:- A trigger is a PL/SQL block that automatically executes when an Insert, UPDATE, DELETE operation occurs/perform on Table.

Triggers are use for auditing, validation, enforcing business rules and automating database actions.

- ★ Why use the triggers:-
- (i). Automatically maintain audit logs.
 - (ii). Validate data.
 - (iii). To Apply business rules.

- (iv). Automatically update related tables.
 (v). Tracks Changes.

* Types of Triggers:-

- (i). BEFORE Trigger:- Runs before the event.

BEFORE INSERT
 BEFORE UPDATE
 BEFORE DELETE

Ex:-> Validate salary
 before insertion.

- (ii). AFTER Trigger:- Runs after the event.

AFTER INSERT
 AFTER UPDATE
 AFTER DELETE

Ex:-
 Store audit information
 after update.

- (iii) STATEMENT-LEVEL Trigger:- Run once per SQL statement.

Ex:- CREATE OR REPLACE TRIGGER tog-test
 AFTER UPDATE
 ON Employee
 BEGIN
 DBMS_output.put_line ('updated');
 END;

Notice- There is "no" for each row

If this query updates 100 rows:
 UPDATE employee
 SET salary = salary + 1000;

The trigger runs only 1 time.

- (iv) Row-level Trigger:- Runs once for each affected row.

Ex:- CREATE OR REPLACE TRIGGER tog-test
 AFTER UPDATE
 ON employee
 FOR EACH ROW
 BEGIN
 DBMS_output.put_line ('updated');
 END;

Now if:- UPDATE Employee
 SET salary = salary + 1000;

updates 100 rows,
 the trigger runs 100 times (once for each row)

- * OLD and NEW Values:- Used in row-level triggers.

OLD -> Value before change.

Ex:-> old.salary



New → Value After Change

Ex: New Salary

Ex: - CREATE OR REPLACE TRIGGER trg-Salary
BEFORE UPDATE
ON employee
FOR EACH ROW
BEGIN

DBMS-Output.Put-Line(
'old Salary: ' || :old.Salary);

DBMS-Output.Put-Line(
'New Salary: ' || :New.Salary);

END;

* Insert Trigger → Run automatically whenever
a new row is inserted.

Ex: - CREATE OR REPLACE TRIGGER trg-insert-emp
AFTER INSERT
ON employee
FOR EACH ROW
BEGIN

DBMS-Output.Put-Line('New employee insert');

END;



When:-

Insert into employee
Values (101, 'John', 50000);

The trigger fires automatically.

* UPDATE Trigger:- Runs automatically whenever
a row is updated.

CREATE OR REPLACE TRIGGER trg-update-emp
AFTER UPDATE
ON employee
FOR EACH ROW
BEGIN

DBMS-Output.Put-Line('Employee record
updated');

END;

When:-

UPDATE employee
SET salary = 60000
Where emp-id = 101;

The trigger fires automatically.